CHAPTER 02

The Sun, Earth & Moon

The Moon looks different every night. Some days are long and warm, others short and cold. Once or twice a year, the sky does something amazing during an eclipse. All of these happen because of how the Sun, Earth, and Moon move together.

📚 Georgia Standard S6E2

WHAT YOU'LL LEARN

- **01** The Sun, Earth & Moon
- **02** Why the Moon Has Phases
- **03** The Eight Moon Phases
- **04** Solar & Lunar Eclipses
- **05** Why We Have Seasons
- **06** Tilt, Not Distance

2.1 The Sun, Earth & Moon

Three objects in space shape our daily lives. The **Sun** is a giant star that gives us light and heat. The **Earth** is our home planet. The **Moon** is Earth's natural satellite — a rocky object that orbits us. Understanding how they move explains day and night, the phases of the Moon, eclipses, and the seasons.

Three Important Motions

Motion	What Moves	How Long It Takes	What It Causes
Earth rotates	Earth spins on its axis	1 day (24 hours)	Day and night
Moon orbits Earth	Moon travels around Earth	About 1 month	The phases of the Moon
Earth orbits Sun	Earth travels around the Sun	1 year (365¼ days)	The seasons

> **Satellite:** any object that orbits a larger object in space. The Moon is Earth's natural satellite. The GPS and TV satellites people launch are artificial satellites.

? Quick Check 2.1

1. What motion causes day and night?
2. About how long does it take the Moon to orbit Earth once?

2.2 Why the Moon Has Phases

Here is something surprising: the Moon does not make its own light. It is like a giant mirror — we only see the part of the Moon that the **Sun is lighting up**.

The Sun always lights up exactly **one half** of the Moon. But as the Moon orbits Earth, we see that lit half from different angles. Sometimes we see all of the lit side, sometimes only a sliver, and sometimes none of it. The shape we see from Earth is called a **phase**.

★ The Key Idea

The Moon's shape doesn't actually change — it's always a round ball. What changes is **how much of the lit half we can see** from Earth as the Moon moves around us. Phases are about our point of view.

! Common Mix-Up

Phases are NOT caused by Earth's shadow. A lot of people think the dark part of the Moon is Earth's shadow falling on it — but that's not it. The dark part is simply the half of the Moon the Sun isn't lighting up at that angle. (Earth's shadow only matters during a lunar eclipse, which is different.)

Quick Check 2.2

1. Where does the Moon's light come from?
2. True or false: the Moon changes shape as it orbits Earth. Explain your answer.

2.3 The Eight Moon Phases

As the Moon orbits Earth over about a month, it goes through eight phases in order. They follow a smooth pattern: from no light (new moon) to fully lit (full moon) and back again.

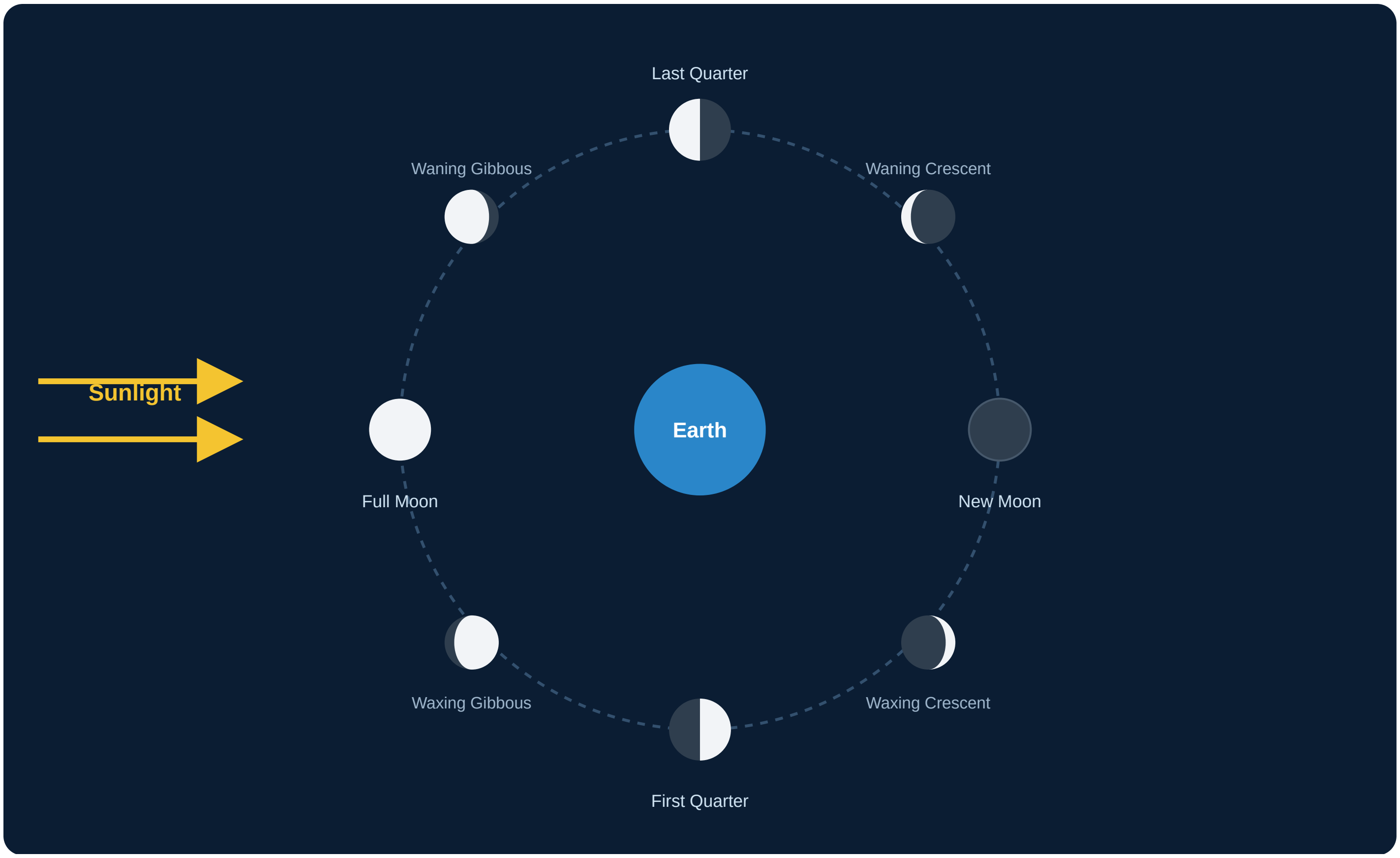

Figure 2.1 — The Moon orbits Earth. The Sun (left) always lights the same half. The labels show the phase we see from Earth at each position.

Waxing vs. Waning

Two words describe whether the lit part is growing or shrinking:

Word	Meaning	What's Happening
Waxing	Getting bigger	The lit part grows from new moon toward full moon
Waning	Getting smaller	The lit part shrinks from full moon back toward new moon
Crescent	A sliver	Less than half of the Moon is lit
Gibbous	More than half	More than half (but not all) of the Moon is lit

★ Memory Trick

Waxing means the light is "**wax**ing on" — growing. **Wan**ing sounds like "**wan**e away" — shrinking. In the Northern Hemisphere, if the **right** side is lit, it's waxing (growing). If the **left** side is lit, it's waning (shrinking).

? Quick Check 2.3

1. Put these in order starting from new moon: full moon, first quarter, new moon, last quarter.
2. What is the difference between a crescent and a gibbous moon?

2.4 Solar & Lunar Eclipses

An **eclipse** happens when the Sun, Earth, and Moon line up in a straight row, so one object's shadow falls on another. There are two kinds, and the order of the three objects tells you which is which.

Solar Eclipse — the Moon blocks the Sun

During a **solar eclipse**, the **Moon** moves directly between the Sun and Earth. The Moon's shadow falls on Earth, and for a short time the daytime sky goes dark. **Order: Sun → Moon → Earth.**

Figure 2.2 — In a solar eclipse, the Moon is in the middle and casts its shadow onto Earth.

Lunar Eclipse — Earth blocks the Sun's light to the Moon

During a **lunar eclipse**, the **Earth** moves directly between the Sun and the Moon. Earth's shadow falls on the Moon, which often turns a dim reddish color. **Order: Sun → Earth → Moon.**

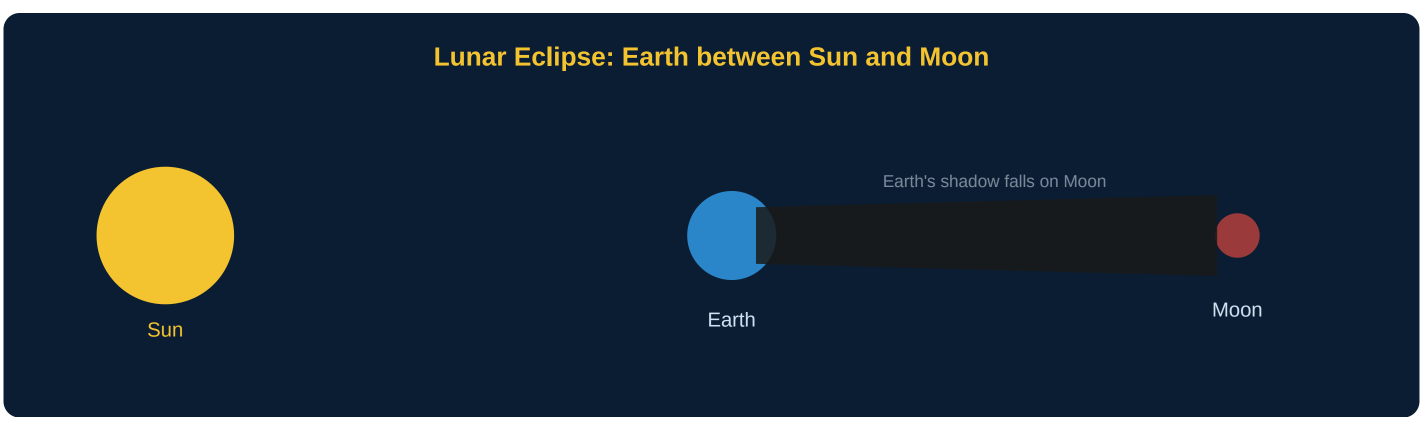

Figure 2.3 — In a lunar eclipse, Earth is in the middle and casts its shadow onto the Moon.

★ How to Keep Them Straight

Name the eclipse after the object that gets **blocked or darkened**. In a **solar** eclipse, the **Sun** gets blocked. In a **lunar** eclipse, the **Moon** gets darkened. Also: the object in the **middle** is the one casting the shadow.

! Why Not Every Month?

The Moon orbits Earth every month, so why don't we get eclipses every month? Because the Moon's orbit is slightly **tilted**. Most months the shadow misses, passing a little above or below. Eclipses only happen when everything lines up just right.

? Quick Check 2.4

1. In a solar eclipse, which object is in the middle?
2. During which eclipse does Earth's shadow fall on the Moon?

2.5 Why We Have Seasons

Many people think summer happens because Earth is closer to the Sun, and winter because it's farther away. That sounds reasonable — but it's **wrong**. The real reason is Earth's **tilt**.

Earth is tilted on its **axis** by about 23.5 degrees, like a slightly leaning spinning top. As Earth orbits the Sun, this tilt stays pointed in the same direction. So during part of the year the Northern Hemisphere leans **toward** the Sun, and during another part it leans **away**.

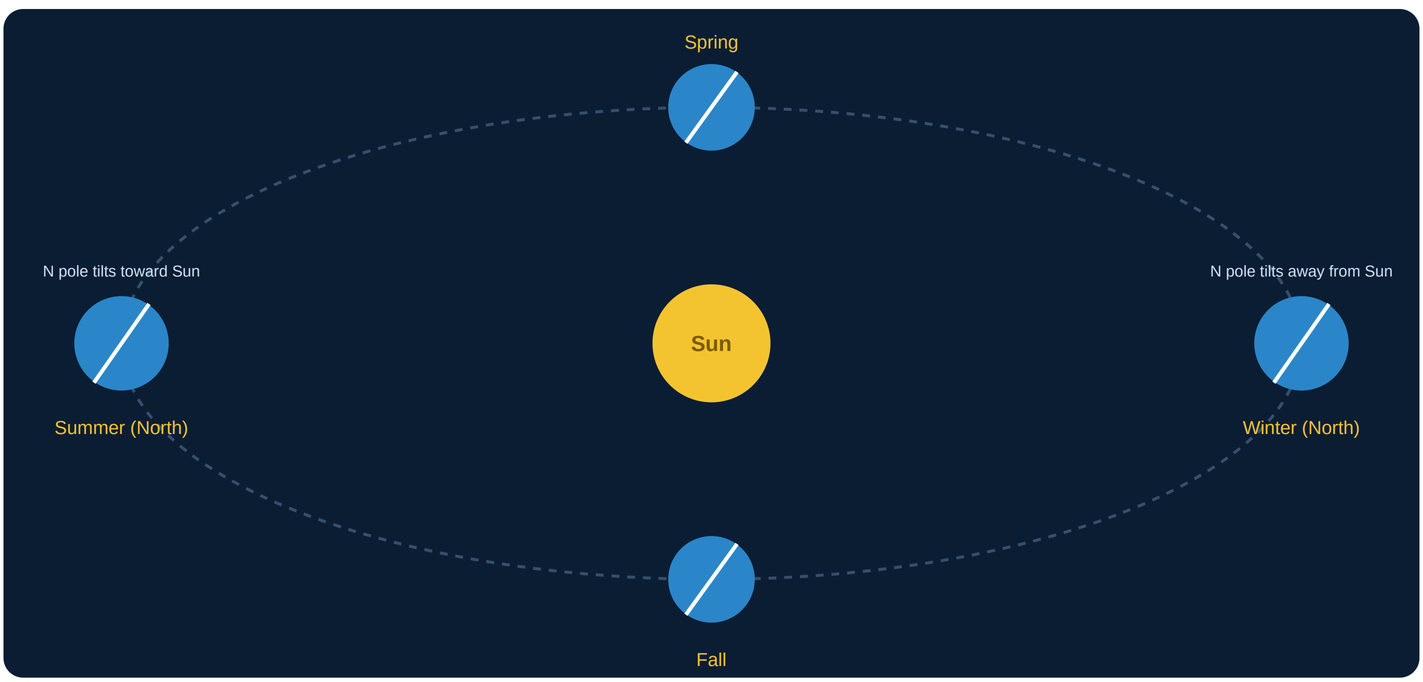

Figure 2.4 — Earth's tilt stays pointed the same way all year. When your hemisphere tilts toward the Sun, you get summer; when it tilts away, you get winter.

What the Tilt Does

When your hemisphere...	Sunlight is...	Days are...	Season
Tilts toward the Sun	Direct and concentrated	Longer	Summer
Tilts away from the Sun	Spread out and weak	Shorter	Winter

When sunlight hits the ground **directly** (straight down), its energy is concentrated and heating is strong. When sunlight hits at a **low angle** (slanted), the same energy spreads over more ground, so heating is weaker. That's why summer is hotter even though the Sun is the same distance away.

> **? Quick Check 2.5**
>
> 1. What is the real cause of the seasons?
> 2. Why is direct sunlight warmer than slanted sunlight?

2.6 Tilt, Not Distance

Here's the proof that distance doesn't cause the seasons: when it's **summer** in the Northern Hemisphere, it's **winter** in the Southern Hemisphere — at the very same time.

If distance from the Sun controlled the seasons, the whole planet would have summer or winter together. But it doesn't. The two hemispheres have **opposite** seasons because they tilt toward the Sun at opposite times of year.

Time of Year	Northern Hemisphere	Southern Hemisphere
June	Summer (tilted toward Sun)	Winter (tilted away)
December	Winter (tilted away)	Summer (tilted toward Sun)

> **★ Surprising Fact**
>
> Earth is actually **closest** to the Sun in early January — during Northern Hemisphere winter! This is more proof that the tilt, not the distance, is what makes the seasons.

> **? Quick Check 2.6**
>
> 1. When it's summer in Georgia, what season is it in Australia? Why?
> 2. If Earth were not tilted, would we still have seasons? Explain.

Chapter 2 Wrap-Up

★ Key Takeaways

- **Three motions** matter: Earth's spin (day/night), the Moon's orbit (phases), and Earth's orbit (seasons).
- **Moon phases** happen because we see different amounts of the Sun-lit half of the Moon — not because of Earth's shadow.
- **Eight phases** cycle from new to full and back; waxing means growing, waning means shrinking.
- **Solar eclipse:** Moon in the middle (Sun→Moon→Earth). **Lunar eclipse:** Earth in the middle (Sun→Earth→Moon).
- **Seasons** are caused by Earth's 23.5° tilt, not its distance from the Sun.